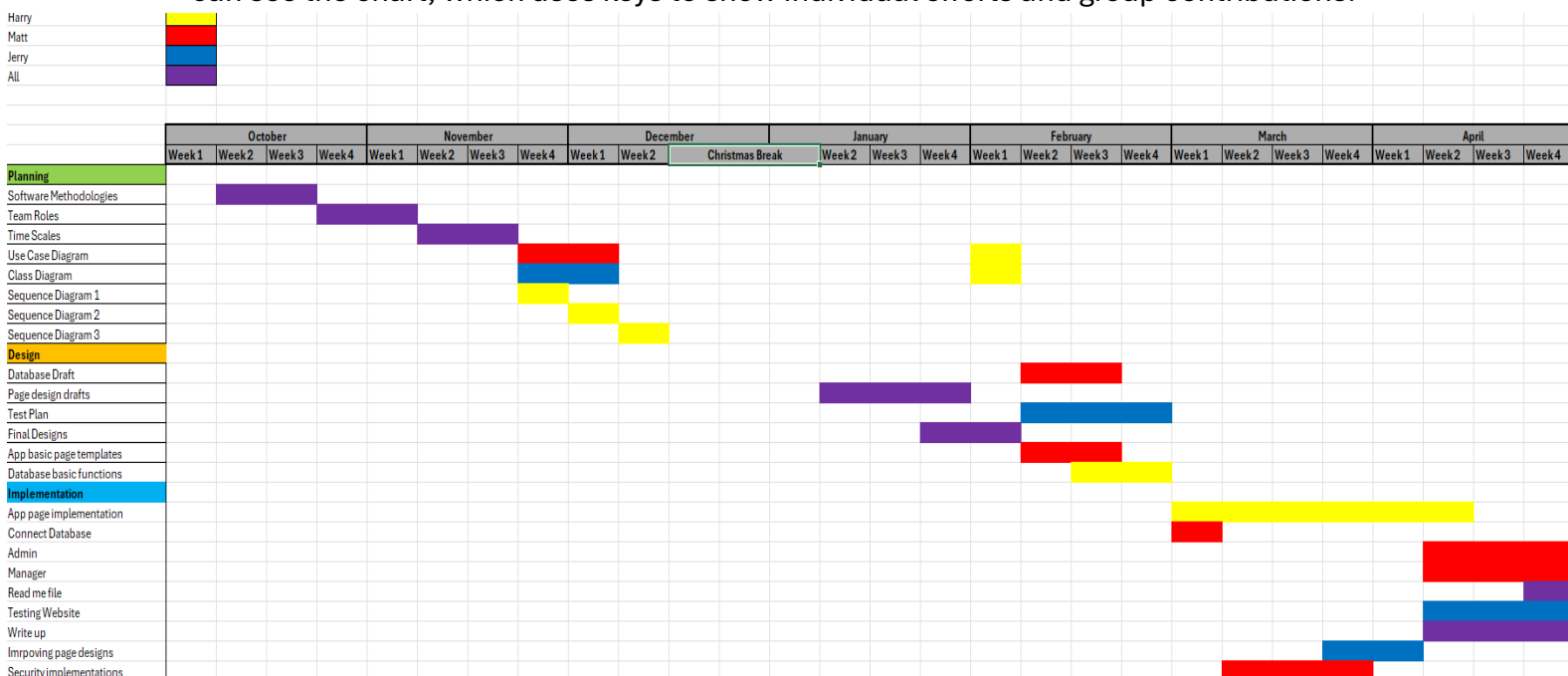


Methodologies and Individual Roles

When developing the project plan and timescale, we decided that using a gantt chart would be best as it is very proficient at visualizing timelines and participation. Below you can see the chart, which uses keys to show individual efforts and group contributions.



We will use the Agile software development methodology as we believe it is a good idea to strive to improve on iterations of the program and get a base model of the app out as soon as possible. This is referred to as a minimum viable product and will implement our core features. Using an iterative methodology will help mitigate the risks as you can identify, and then later solve errors that occur in the development process. For instance, the cinema may have flaws in its database, which would be identified in testing, and solved in the later iterations' development. The methods scalability and flexibility make it easier to add additional criteria as we see fit like security, for instance. This also relies on us control versions correctly, as constant updates might lead to mergers and clashes on git. To prevent this, we will pull any changes before beginning our own development and update the group when changes have been made. Our Agile workflow included weekly sprints with planning meetings every Friday to assign and review tasks. We maintained a shared backlog and used GitHub repository to track progress and dependencies.

The group roles have been split up based on individual strengths as well as what we would enjoy developing on, this is important for motivation and to make sure that each member is qualified to complete their works fully. Individual roles are displayed on the chart above and more in detail below.

Individual roles really began to develop in the design and implementation section, where we split up what each person would be in charge of.

Harry's Role:

- Develop the main user experience of the app.
- Built out the film listings: see all film details (description, actors, genre ratings).
- Complete the booking process: seat management working, calculate ticket pricing correctly (depending on location, time, and seat type), and generate unique booking codes).
- User can view their own booking (alongside key details).
- Code the cancellation rules: 50% fee, cancel on day of not allowed.

Matt's Role:

- Design login experience and page templates. Smooth sign up and log in process.
- Passwords stored safely (hashed) and users only see what they are supposed to.
- Data validation so no bad data gets into the system.
- Develop management and admin side of the system.

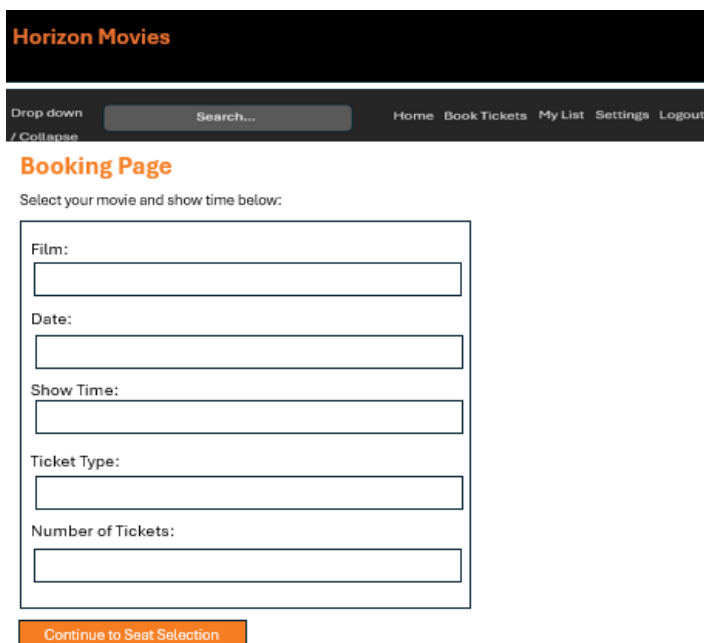
Jerry's Role:

- Polish the look and feel of the app. Increase professionalism, make everything user-friendly, across all pages.
- Ensure everything works as intended.
- Add features like current movies and posters.
- Test planning and testing.
- Add commenting and documenting.

Evaluation:

As a group we have worked well to communicate and develop a product. We have delivered a product which meets the specification very well and have done so whilst largely sticking to timescales and plans. The decision to take some time to design the base pages have saved time in developing from scratch as we could refer back to what we initially envisioned for the pages. Seen below.

Figure 1 - Booking Page Design



Horizon Movies

Drop down Search... Home Book Tickets My List Settings Logout / Collapse

Booking Page

Select your movie and show time below:

Film:

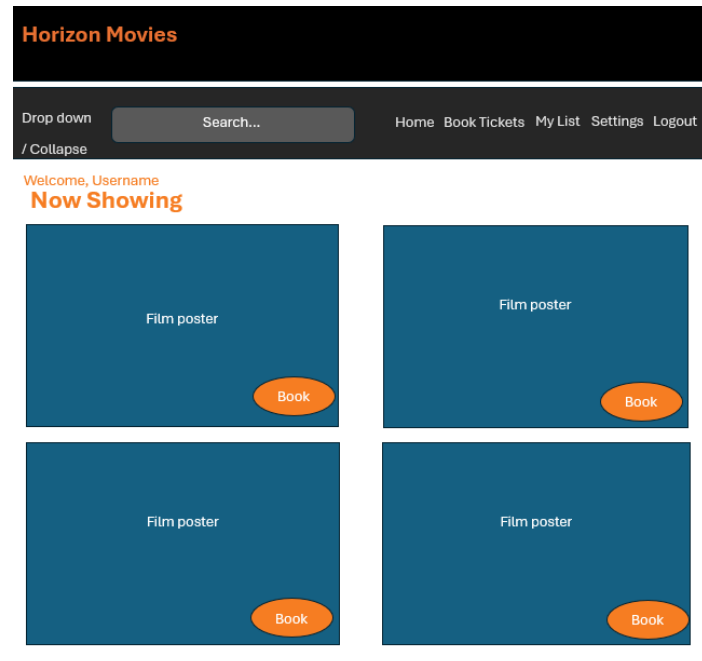
Date:

Show Time:

Ticket Type:

Number of Tickets:

[Continue to Seat Selection](#)



Horizon Movies

Drop down Search... Home Book Tickets My List Settings Logout / Collapse

Welcome, Username

Now Showing

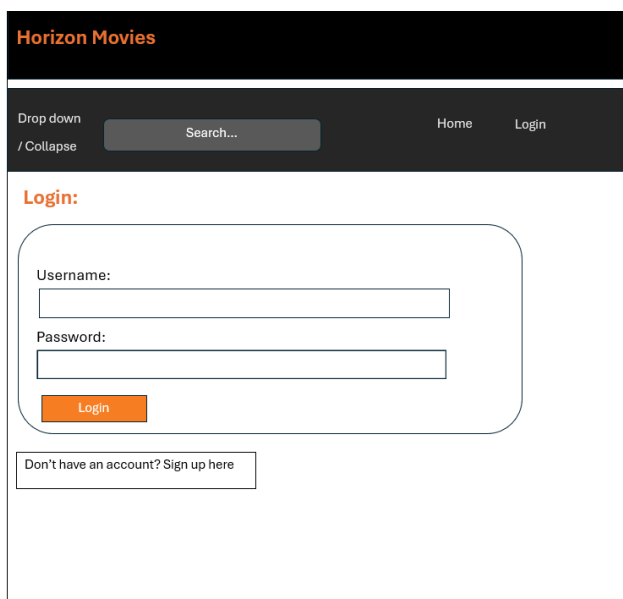
Film poster [Book](#)

Film poster [Book](#)

Film poster [Book](#)

Film poster [Book](#)

Figure 2 - Home page design 1



Horizon Movies

Drop down Search... Home Login / Collapse

Login:

Username:

Password:

[Login](#)

[Don't have an account? Sign up here](#)

Figure 3 - login page design 1

Additionally, all of the diagrams helped our understanding of the project as a whole, and what to expect in terms of developing the database, developing pages to fit standards, and expected errors. As far as the base application, there could be some visual improvements, as it still looks very much like a prototype, but most importantly everything works as it should due to thorough testing and implementation. Additionally, we agree that as a group we worked well together, roles we split evenly between the

members, and we stayed in contact, which meant we were always ready to message or reply to questions from other members if they were unsure. Thankfully, due to consistent coding styles and correct variable naming, alongside commenting, made it very easy to pick up the project at any time and add onto it.